Neural Layout Analysis for Voice Narration

Speech narration in digital reading systems requires extracting main body content while cleanly skipping tabular data, captions, and running page headers.

Benchmark Evaluation

We evaluate deep parsing models across structured academic articles and reports.

Architecture	Precision	Recall	F1 Score
Baseline Parser	82.4%	79.1%	80.7%
Docling Hybrid	96.8%	95.4%	96.1%

Table 1: Quantitative evaluation comparing baseline parser with Docling hybrid layout analysis.

The experimental results demonstrate that neural layout analysis prevents unnatural TTS reading of table cells and captions.

Parameter	Configuration Value	Hardware Target
Batch Size	32 samples per device	NVIDIA RTX 4090
Learning Rate	1e-4 with warmup	Distributed DDP
Dropout Rate	0.15 regularized	Float16 TensorCore
Quantization	8-bit integer weights	ONNX Runtime CPU

Table 2: Supplementary hyperparameters and target deployment hardware configurations.

Conclusion and Future Work

In summary, integrating deep layout models with speech synthesis ensures a seamless listening experience.

Future work will extend boundary detection to multi-column complex financial spreadsheets and scanned archives.